## GRAPH AND KNOWLEDGE ANALYTICS INTERN

# CHALLENGE 2 — LOG ANALYSIS

*To Ascertain Who Is Compromised and How*

# 1. Personal Details

| | |
|---|---|
| **Name** | Jitesh Kumar |
| **Course** | B.Tech (Computer Science and Engineering) |
| **Year / Semester** | 4th Year / 7th Semester |
| **College** | Lloyd Institute of Engineering and Technology |
| **University** | Dr. A.P.J. Abdul Kalam Technical University, Lucknow |
| **University Roll No.** | 2301530100066 |
| **Email / Mobile** | Jiteshkumarbtech2327@liet.in \| 9027444840 |

# 2. Executive Summary

The investigation analyzed endpoint web activity, endpoint security events and USB activity to identify endpoints associated with a common malicious artifact and assess the available evidence for its possible access vector.

A common malicious SHA-256 IOC, 5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c, corresponding to the filename recently-used.xbel, was observed across exactly three endpoints: EPJYEC7F, EPOC6Y4R and EPC2L3UH. The IOC generated 37 observed events across these endpoints.

Each endpoint's first observed IOC detection was accompanied by an immediate quarantine event at the same timestamp (0 seconds), indicating immediate security-control remediation.

No matching web activity was found within ±2 hours and no matching USB activity was found within ±30 minutes of the first IOC detection for the three primary endpoints. Therefore, the available telemetry does not establish the initial access vector.

The evidence supports association with/impact from the same malicious artifact and immediate quarantine. It does not establish execution, persistence, initial access mechanism, or endpoint-to-endpoint propagation.

# 3. Tools Used

- Python
- Jupyter Notebook
- Pandas — data loading, filtering, grouping and correlation
- Regular Expressions (re) — text/IOC handling where required
- Datetime / timestamp handling — temporal correlation

NetworkX and Matplotlib are included only for the final legitimate graph-analysis layer, provided that the graph is actually generated from the verified relationships.

# 4. Methodology

## 4.1 Analysis

The investigation followed a question-driven forensic workflow: understand the datasets, prepare timestamps, identify suspicious/malicious events, extract the common IOC, map the IOC to endpoints, construct timelines, test web and USB correlations, validate quarantine/remediation, and finally distinguish supported findings from unsupported inferences.

Primary questions:

| Question | Investigation Objective |
|---|---|

| | |
|---|---|
| **WHO?** | Which endpoints are associated with the malicious IOC? |
| **WHAT?** | What exact artifact/IOC is involved? |
| **WHEN?** | When was it first and last observed? |
| **HOW?** | Can the available telemetry establish the initial access vector? |
| **PROPAGATION?** | Can endpoint-to-endpoint spread be established? |
| **REMEDIATION?** | Was the artifact quarantined? |

## 4.2 Finding

Core IOC and endpoint mapping:

| Endpoint | Host | Unit | IOC Events | First Observed | Last Observed |
|---|---|---|---|---|---|
| EPJYEC7F | host-38aa | unit-romeo | 10 | 2026-09-01 08:09:22 | 2026-09-26 06:45:13 |
| EPOC6Y4R | host-e783 | unit-charlie | 9 | 2026-09-01 08:59:34 | 2026-09-24 08:39:34 |
| EPC2L3UH | host-b4da | unit-india | 18 | 2026-09-02 08:48:22 | 2026-09-23 08:45:44 |

**Common SHA-256:**
5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c

**Filename:** recently-used.xbel

**Total observed IOC events:** 37

### First detection and quarantine

| Endpoint | First IOC Detection | Quarantine | Delta |
|---|---|---|---|
| EPJYEC7F | 2026-09-01 08:09:22 | Same timestamp | 0 sec |
| EPOC6Y4R | 2026-09-01 08:59:34 | Same timestamp | 0 sec |
| EPC2L3UH | 2026-09-02 08:48:22 | Same timestamp | 0 sec |

Interpretation: the security control detected the artifact and immediately quarantined it. This establishes detection/remediation, not malware execution, persistence, infection origin or propagation.

## Web and USB correlation

| Endpoint | Web ±2h | USB ±30m | Interpretation |
|---|---|---|---|
| EPJYEC7F | 0 | 0 | Initial access vector not established |
| EPOC6Y4R | 0 | 0 | Initial access vector not established |
| EPC2L3UH | 0 | 0 | Initial access vector not established |

Correct forensic wording: available web telemetry did not establish a web-based initial access vector in the tested window; available USB telemetry did not establish USB as the initial access vector in the tested window.

## Additional endpoint findings

EP4YIFHV: 9 web/DNS records at 09:02:52 were followed 20m41s later by 20 distinct malicious files quarantined at 09:23:33 and the same 20 files detected/quarantined again at 09:35:09. The two file clusters had the same filename set and SHA-256 set. This supports repeat detection of the same artifact set; the exact trigger for the second detection was not established.

EPPGWIOF: 10 malicious events represented 2 files repeated across 5 timestamps: brprintconflsr3 and rawtobr3. Therefore, 10 events are not 10 unique malicious files. No web activity or AV event was found within the tested ±2-hour window around first detection.

EPD24DBC: USB mass-storage activity was observed around 2026-09-22 18:23:12 and 18:23:41, but no security, web or malicious-file events were found within the tested ±30-minute correlation window. USB activity alone is not sufficient to classify this endpoint as compromised.

## 4.3 Outcome

Three endpoints — EPJYEC7F (unit-romeo), EPOC6Y4R (unit-charlie), and EPC2L3UH (unit-india) — are associated with the same malicious SHA-256 IOC 5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c, corresponding to recently-used.xbel. The IOC generated 37 observed events across the three endpoints.

The earliest observed detection was EPJYEC7F, followed temporally by EPOC6Y4R and then EPC2L3UH. This sequence is only a temporal sequence and is not evidence of endpoint-to-endpoint propagation.

All three endpoints had immediate quarantine at first IOC detection. Available web and USB telemetry did not establish the initial access vector. Therefore, the evidence is insufficient to determine how the artifact initially entered the environment or whether one endpoint infected another.

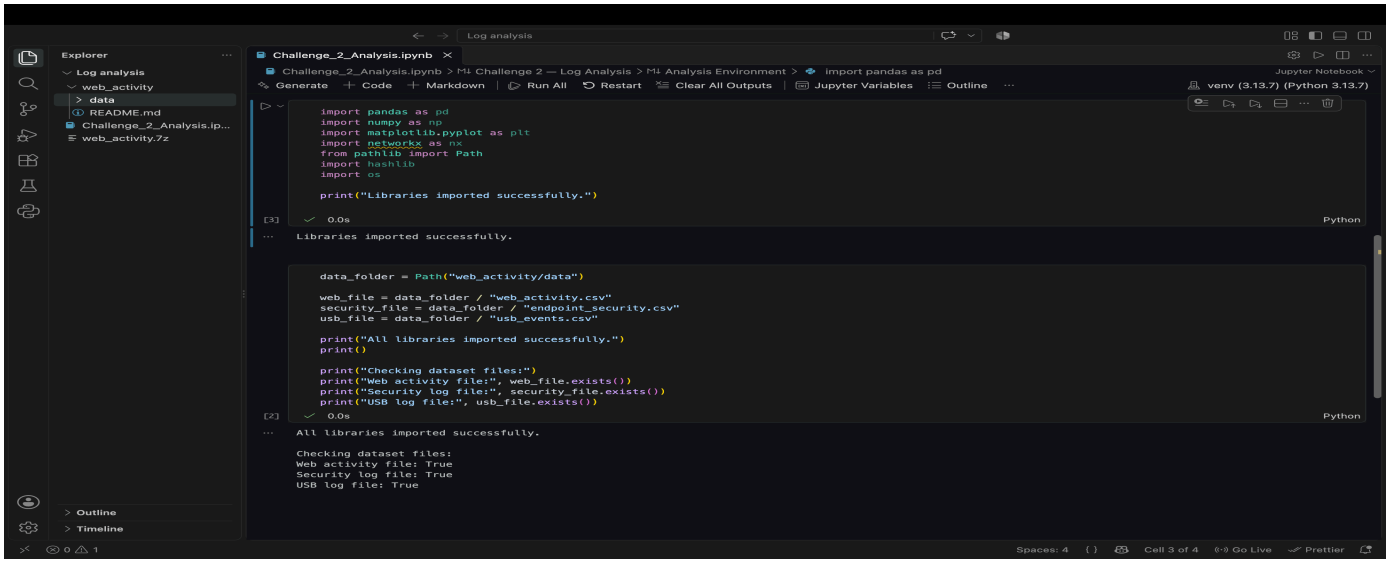
# 5. Step-by-Step Process with Screenshots

The following placeholders are intentionally left for screenshots. Attach only the most evidential notebook outputs; do not attach every cell. Each placeholder states exactly what should be captured.

## Step 1 — Dataset identification

Purpose: Establish what telemetry is available before analysis.

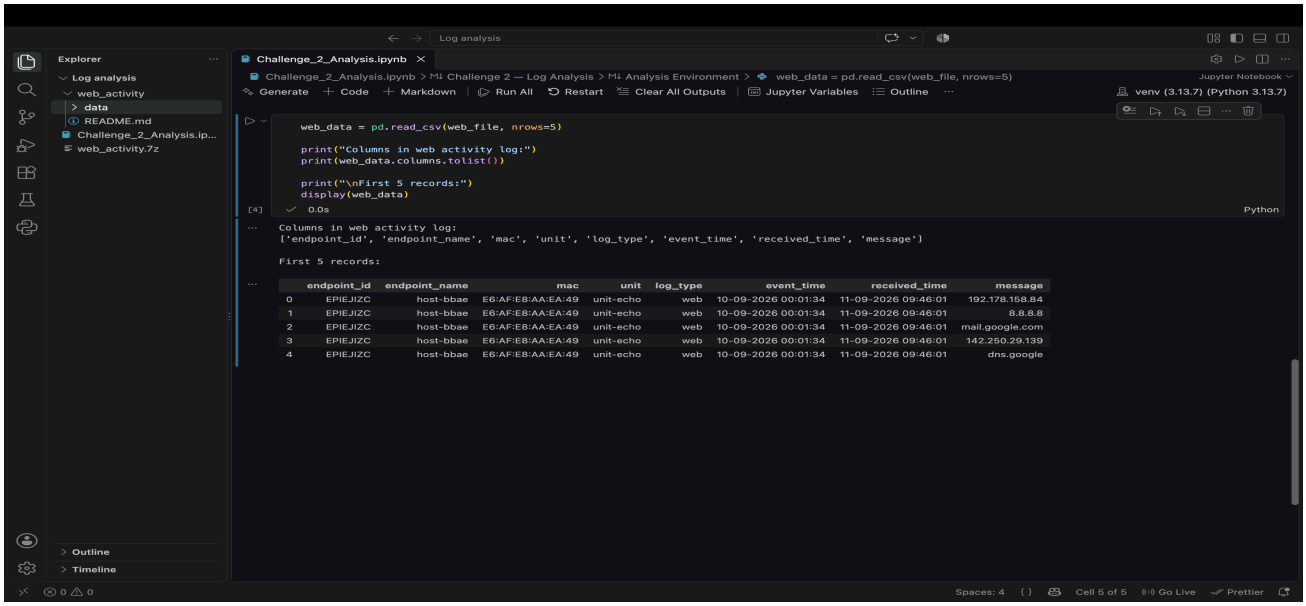
**SCREENSHOT PLACEHOLDER — Step 1 — Dataset identification**



## Step 2 — Dataset inspection

Purpose: Understand schema and coverage.

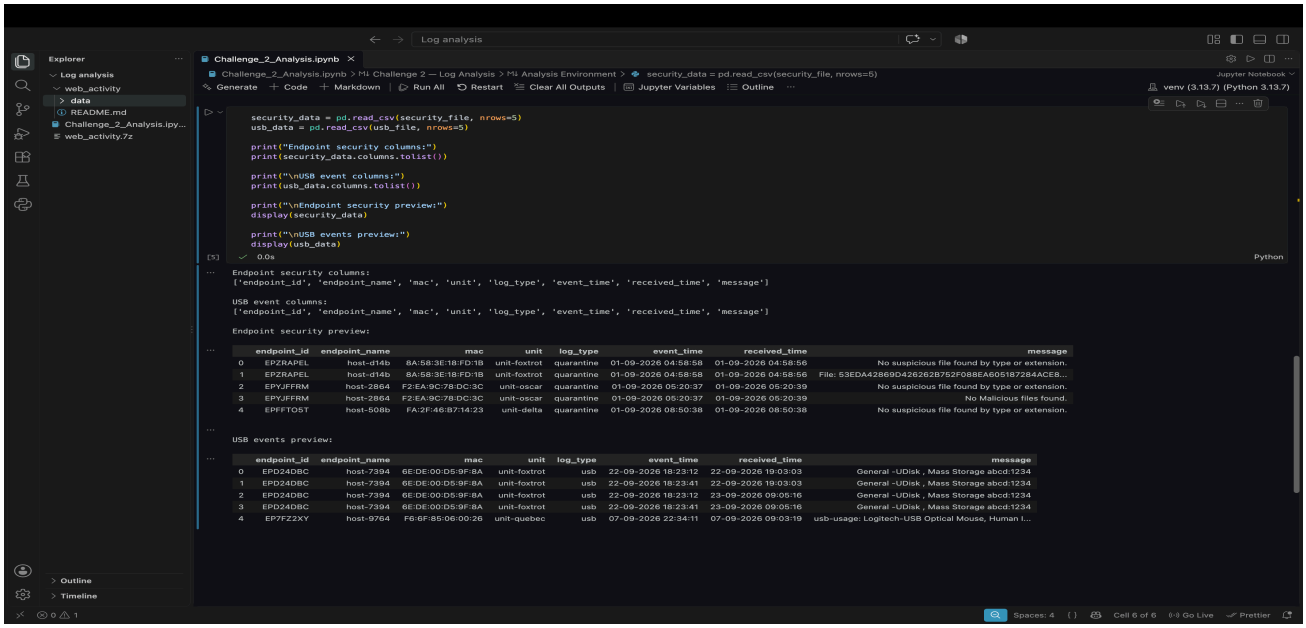
**SCREENSHOT PLACEHOLDER — Step 2 — Dataset inspection**




## Step 3 — Timestamp/data preparation

Purpose: Enable reliable temporal correlation.

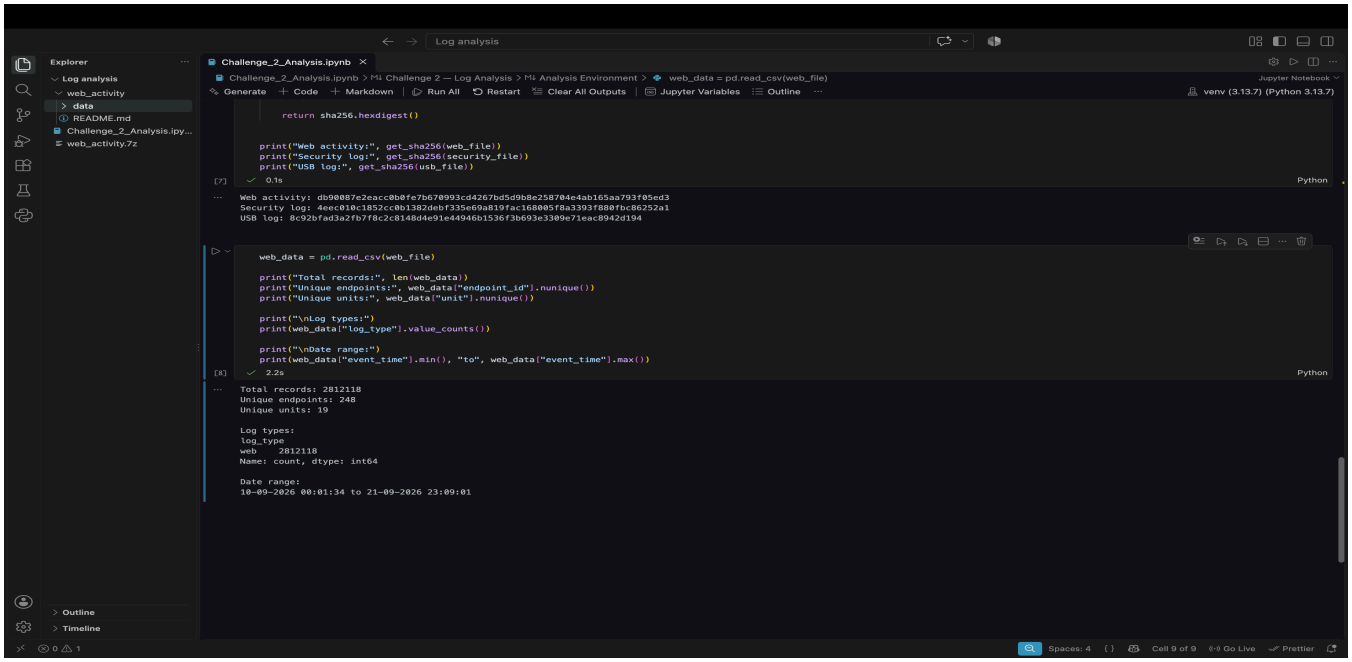
**SCREENSHOT PLACEHOLDER — Step 3 — Timestamp/data preparation**



## Step 4 — Malicious-event identification

Purpose: Reduce the investigation to evidence-bearing events.

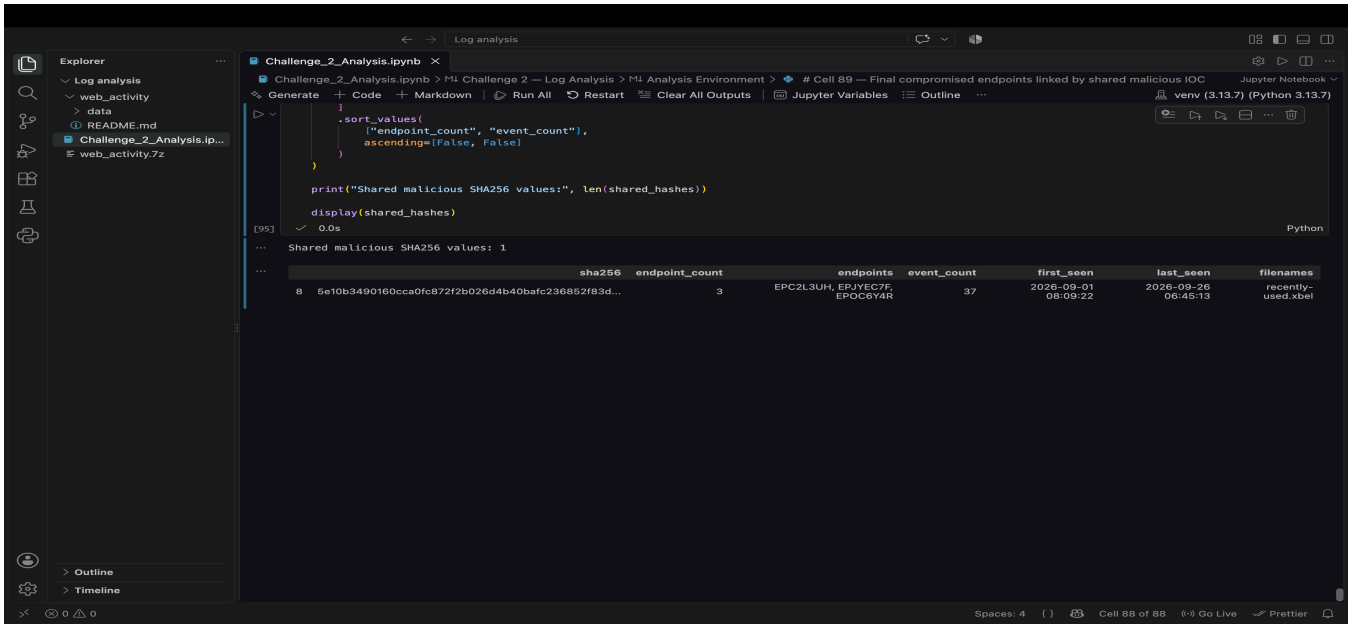
**SCREENSHOT PLACEHOLDER — Step 4 — Malicious-event identification**




## Step 5 — SHA-256 IOC extraction

Purpose: Identify the common indicator linking affected endpoints.

**SCREENSHOT PLACEHOLDER — Step 5 — SHA-256 IOC extraction**



## Step 6 — IOC-to-endpoint mapping

Purpose: Answer WHO is associated with the IOC.

**SCREENSHOT PLACEHOLDER — Step 6 — IOC-to-endpoint mapping**

```
Shared IOC: 5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
Total shared-IOC events: 37

First detection per endpoint:
```

| | endpoint_id | endpoint_name | unit | first_seen | last_seen | event_count |
|---|---|---|---|---|---|---|
| 1 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-01 08:09:22 | 2026-09-26 06:45:13 | 10 |
| 2 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-01 08:59:34 | 2026-09-24 08:39:34 | 9 |
| 0 | EPC2L3UH | host-b4da | unit-india | 2026-09-02 08:48:22 | 2026-09-23 08:45:44 | 18 |

## Step 7 — Timeline construction

Purpose: Establish chronology without assuming causality.

**SCREENSHOT PLACEHOLDER — Step 7 — Timeline construction**

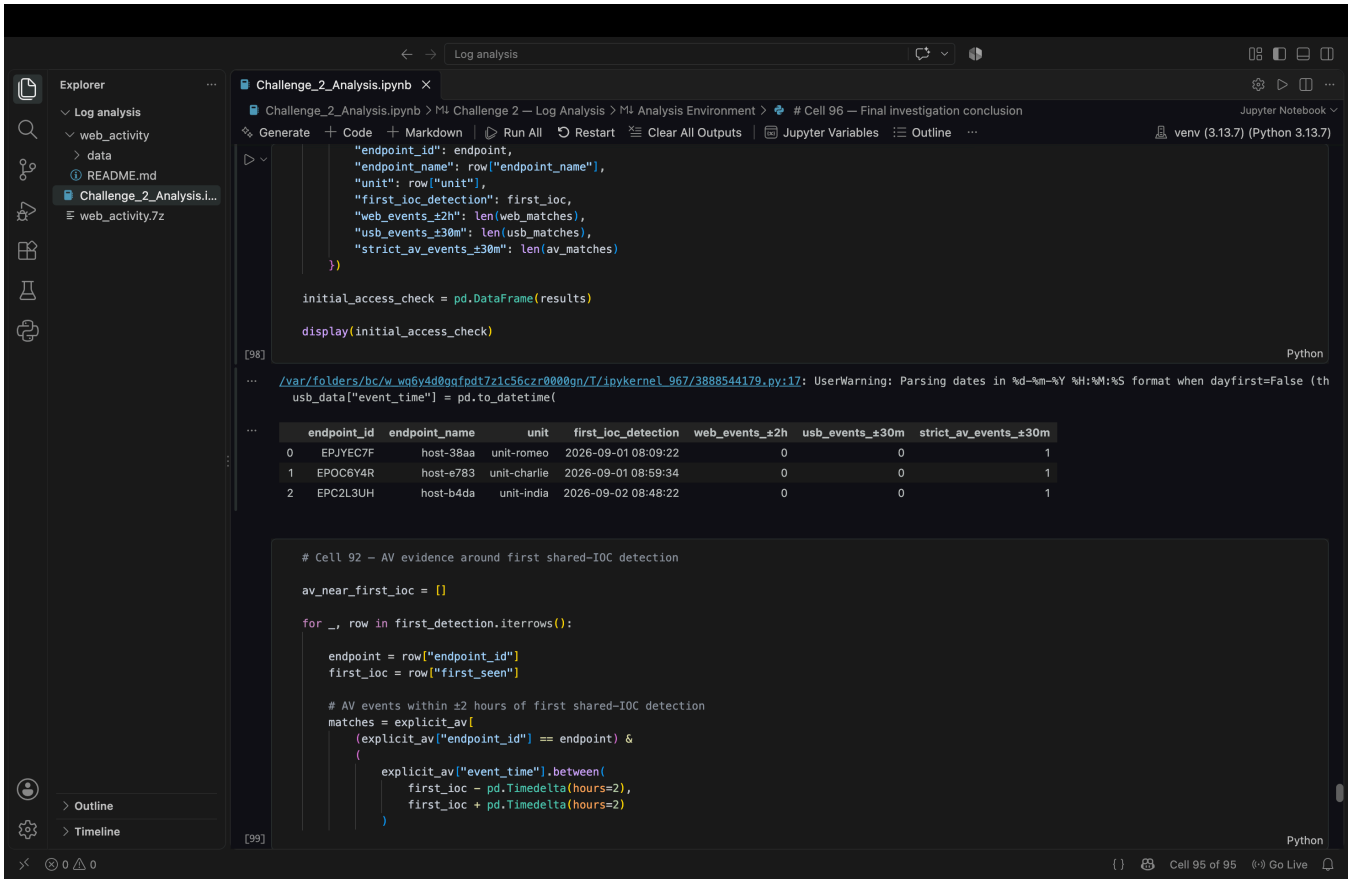
```
Full chronological shared-IOC timeline:
```

| | endpoint_id | endpoint_name | unit | event_time | filename | message |
|---|---|---|---|---|---|---|
| 0 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-01 08:09:22 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 1 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-01 08:59:34 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 2 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-01 18:21:26 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 3 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-02 08:39:53 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 4 | EPC2L3UH | host-b4da | unit-india | 2026-09-02 08:48:22 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 5 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-03 19:14:05 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 6 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-04 08:23:22 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 7 | EPC2L3UH | host-b4da | unit-india | 2026-09-05 08:58:56 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 8 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-07 09:03:50 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 9 | EPC2L3UH | host-b4da | unit-india | 2026-09-08 08:41:56 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 10 | EPC2L3UH | host-b4da | unit-india | 2026-09-09 08:40:46 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 11 | EPC2L3UH | host-b4da | unit-india | 2026-09-09 17:22:33 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 12 | EPC2L3UH | host-b4da | unit-india | 2026-09-10 08:42:46 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 13 | EPC2L3UH | host-b4da | unit-india | 2026-09-12 09:15:26 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 14 | EPC2L3UH | host-b4da | unit-india | 2026-09-14 16:50:33 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 15 | EPC2L3UH | host-b4da | unit-india | 2026-09-15 10:06:02 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 16 | EPC2L3UH | host-b4da | unit-india | 2026-09-15 16:53:38 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 17 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-16 08:20:52 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 18 | EPC2L3UH | host-b4da | unit-india | 2026-09-16 09:53:09 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 19 | EPC2L3UH | host-b4da | unit-india | 2026-09-16 16:55:32 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 20 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-16 17:26:50 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 21 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-17 08:11:38 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 22 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-17 08:26:22 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 23 | EPC2L3UH | host-b4da | unit-india | 2026-09-17 10:17:21 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 24 | EPC2L3UH | host-b4da | unit-india | 2026-09-17 17:08:57 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 25 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-18 08:29:52 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 26 | EPC2L3UH | host-b4da | unit-india | 2026-09-18 09:18:05 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 27 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-18 18:58:47 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 28 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-19 08:20:01 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 29 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-21 08:47:28 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 30 | EPC2L3UH | host-b4da | unit-india | 2026-09-22 09:43:00 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 31 | EPC2L3UH | host-b4da | unit-india | 2026-09-22 13:34:27 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 32 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-23 08:13:14 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 33 | EPC2L3UH | host-b4da | unit-india | 2026-09-23 08:45:44 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 34 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-23 17:48:04 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 35 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-24 08:39:34 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |
| 36 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-26 06:45:13 | recently-used.xbel | File: recently-used.xbel \| Quarantined At: 202... |

## Step 8 — Web correlation

Purpose: Test whether web activity temporally supports initial access.

**SCREENSHOT PLACEHOLDER — Step 8 — Web correlation**




## Step 9 — USB correlation

Purpose: Test whether removable media temporally supports initial access.

**SCREENSHOT PLACEHOLDER — Step 9 — USB correlation**

---Same screenshots listed for step 8-----

## Step 10 — AV/quarantine correlation

Purpose: Validate detection and remediation.

**SCREENSHOT PLACEHOLDER — Step 10 — AV/quarantine correlation**

```
                "endpoint_id",
                "first_ioc_detection",
                "event_time",
                "delta_seconds",
                "log_type",
                "message"
            ]
        ]
    )

else:
    print("NO AV events within ±2 hours of first shared-IOC detection.")
```

```
AV events within ±2 hours of first shared-IOC detection: 3
```

| | endpoint_id | first_ioc_detection | event_time | delta_seconds | log_type | message |
|---|---|---|---|---|---|---|
| 0 | EPC2L3UH | 2026-09-02 08:48:22 | 2026-09-02 08:48:22 | 0.0 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |
| 1 | EPJYEC7F | 2026-09-01 08:09:22 | 2026-09-01 08:09:22 | 0.0 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |
| 2 | EPOC6Y4R | 2026-09-01 08:59:34 | 2026-09-01 08:59:34 | 0.0 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |

```
# Cell 93 — Full quarantine evidence for first shared-IOC detections

first_ioc_quarantine = explicit_av[
    explicit_av["endpoint_id"].isin(first_detection["endpoint_id"])
].copy()

first_ioc_quarantine["delta_seconds"] = None

for _, row in first_detection.iterrows():
```

```
)
```

[101]

```
Infection/remediation events within ±5 minutes: 3
```

| | endpoint_id | first_ioc_detection | event_time | delta_seconds | log_type | message |
|---|---|---|---|---|---|---|
| 0 | EPC2L3UH | 2026-09-02 08:48:22 | 2026-09-02 08:48:22 | 0.0 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |
| 1 | EPJYEC7F | 2026-09-01 08:09:22 | 2026-09-01 08:09:22 | 0.0 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |
| 2 | EPOC6Y4R | 2026-09-01 08:59:34 | 2026-09-01 08:59:34 | 0.0 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |

```
# Cell 95 — Final compromise evidence matrix

final_evidence = first_detection[
    [
        "endpoint_id",
        "endpoint_name"
```

```
display(
    first_ioc_quarantine[
        [
            "endpoint_id",
            "event_time",
            "log_type",
            "message"
        ]
    ].sort_values(["event_time", "endpoint_id"])
)
```

[100]

| | endpoint_id | event_time | log_type | message |
|---|---|---|---|---|
| 995 | EPJYEC7F | 2026-09-01 08:09:22 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |
| 394 | EPOC6Y4R | 2026-09-01 08:59:34 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |
| 1148 | EPC2L3UH | 2026-09-02 08:48:22 | quarantine | File: recently-used.xbel \| Quarantined At: 202... |

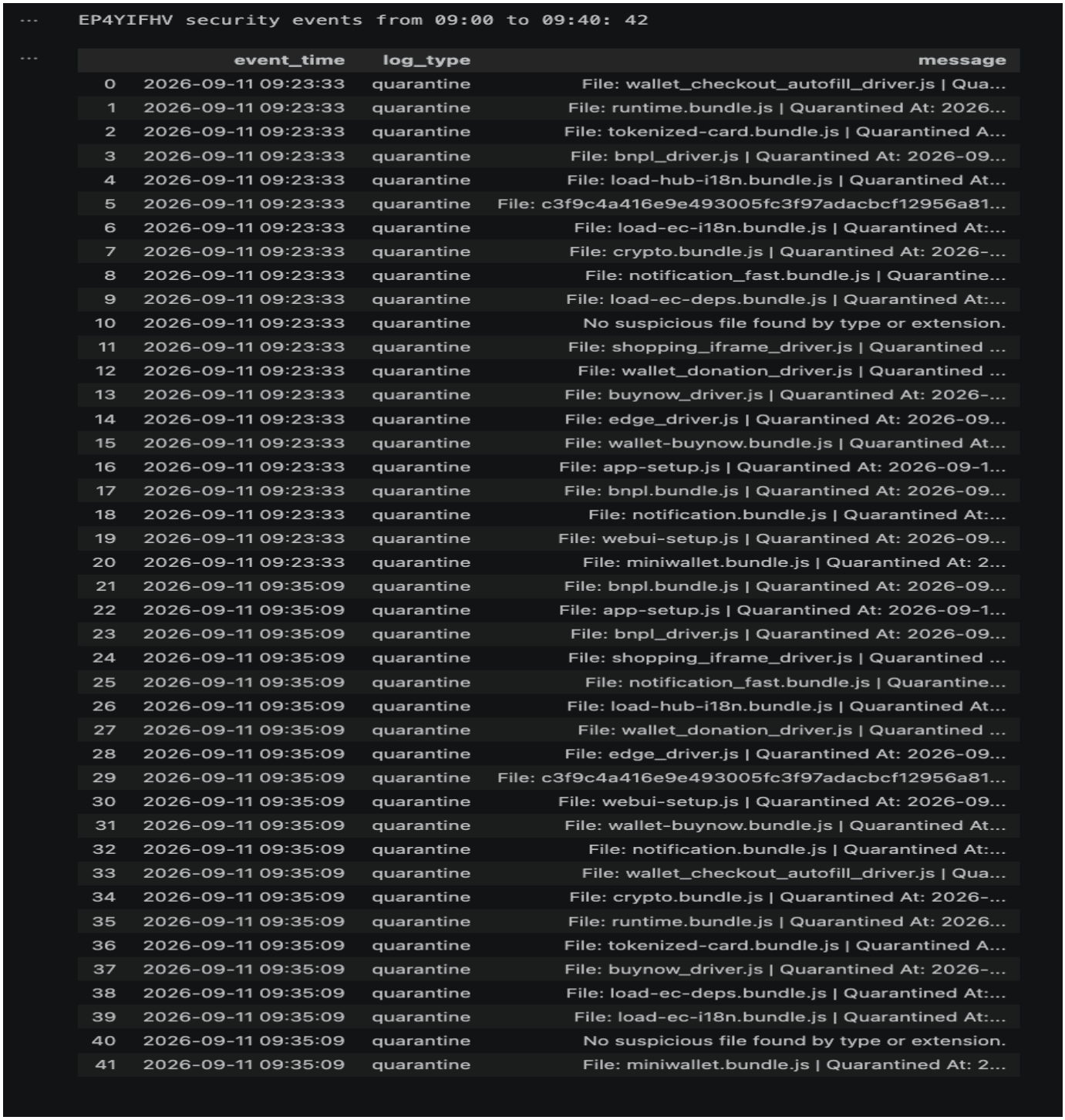
```
# Cell 94 — Check infection/remediation events immediately around first IOC quarantine

infection_keywords = [
    "infected",
    "infection",
    "detected",
    "quarantined",
    "deleted",
    "removed",
```

## Step 11 — Additional endpoint investigations

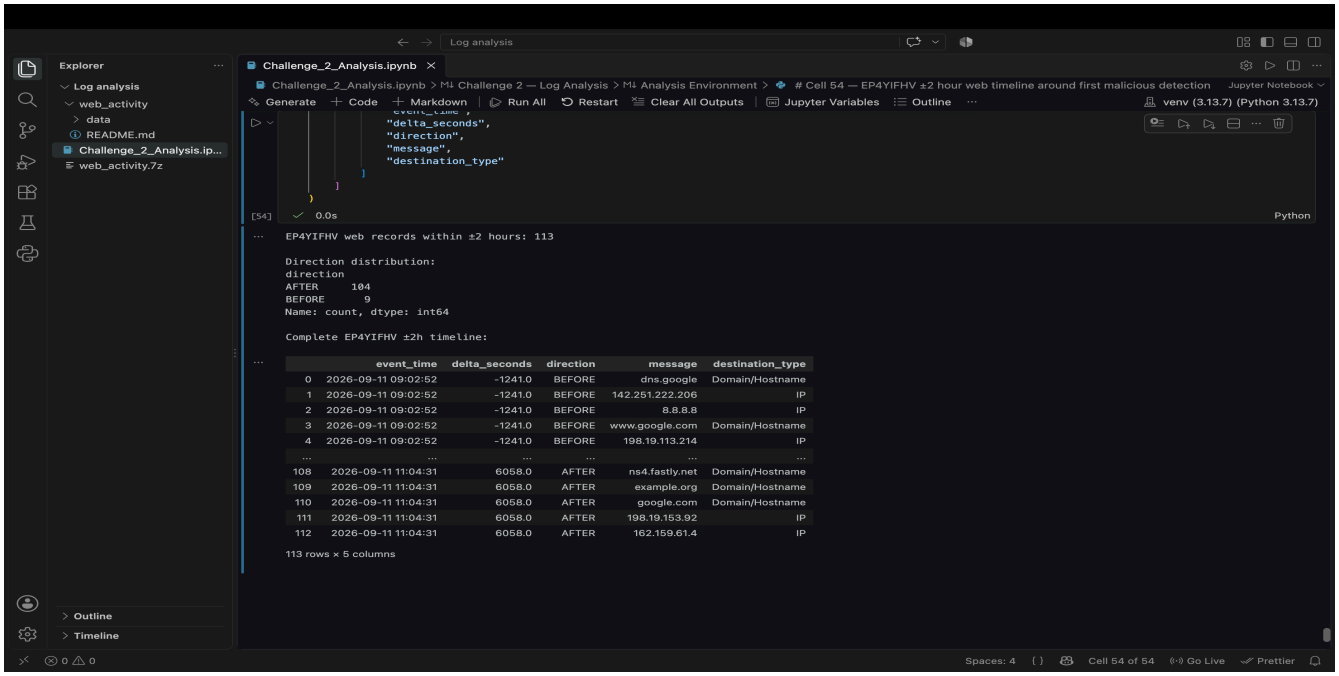
Purpose: Check whether other suspicious patterns are actually supported.

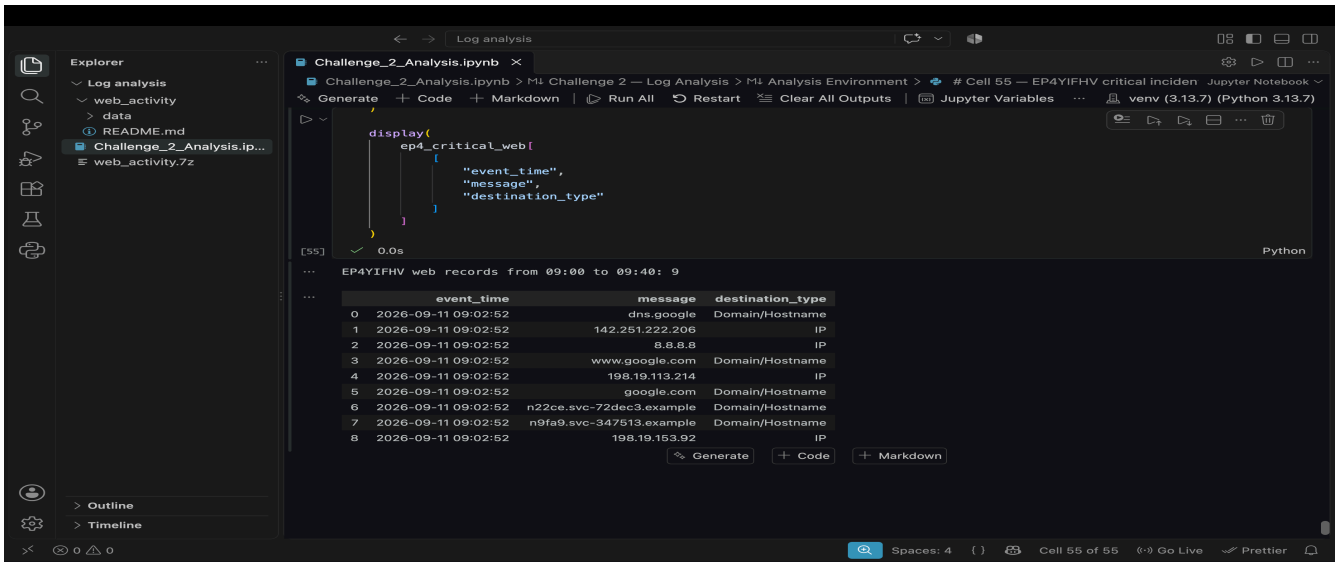
**SCREENSHOT PLACEHOLDER — Step 11 — Additional endpoint investigations**

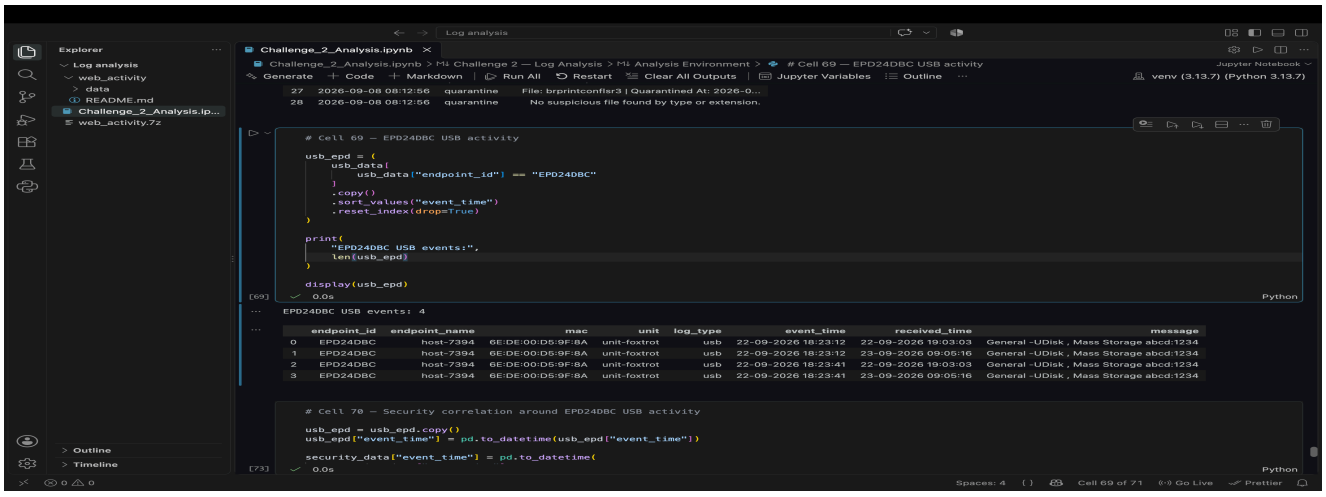


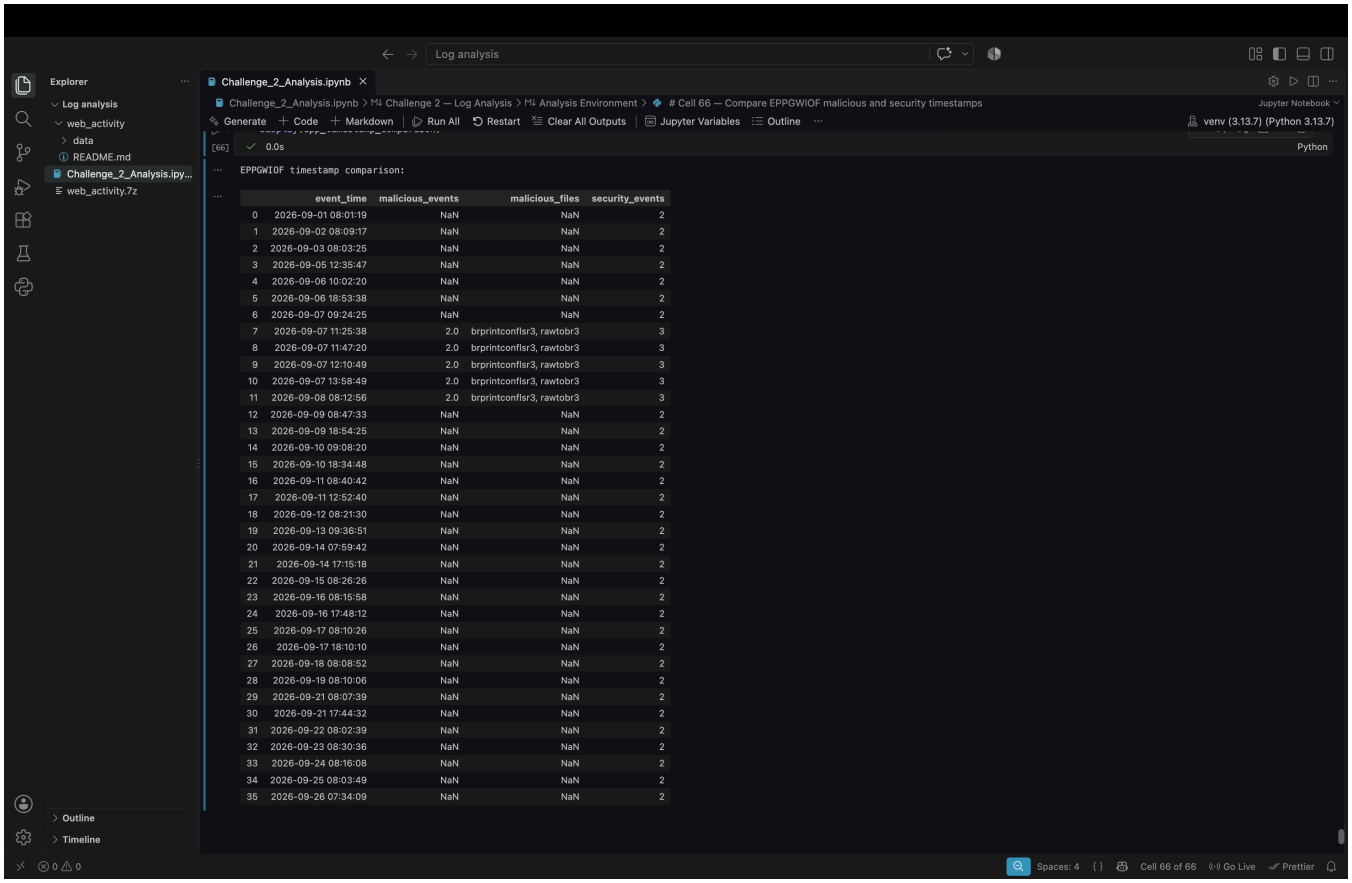
EP4YIFHV security events from 09:00 to 09:40: 42

| | event_time | log_type | message |
|---|---|---|---|
| 0 | 2026-09-11 09:23:33 | quarantine | File: wallet_checkout_autofill_driver.js \| Qua... |
| 1 | 2026-09-11 09:23:33 | quarantine | File: runtime.bundle.js \| Quarantined At: 2026... |
| 2 | 2026-09-11 09:23:33 | quarantine | File: tokenized-card.bundle.js \| Quarantined A... |
| 3 | 2026-09-11 09:23:33 | quarantine | File: bnpl_driver.js \| Quarantined At: 2026-09... |
| 4 | 2026-09-11 09:23:33 | quarantine | File: load-hub-i18n.bundle.js \| Quarantined At... |
| 5 | 2026-09-11 09:23:33 | quarantine | File: c3f9c4a416e9e493005fc3f97adacbcf12956a81... |
| 6 | 2026-09-11 09:23:33 | quarantine | File: load-ec-i18n.bundle.js \| Quarantined At:... |
| 7 | 2026-09-11 09:23:33 | quarantine | File: crypto.bundle.js \| Quarantined At: 2026-... |
| 8 | 2026-09-11 09:23:33 | quarantine | File: notification_fast.bundle.js \| Quarantine... |
| 9 | 2026-09-11 09:23:33 | quarantine | File: load-ec-deps.bundle.js \| Quarantined At:... |
| 10 | 2026-09-11 09:23:33 | quarantine | No suspicious file found by type or extension. |
| 11 | 2026-09-11 09:23:33 | quarantine | File: shopping_iframe_driver.js \| Quarantined ... |
| 12 | 2026-09-11 09:23:33 | quarantine | File: wallet_donation_driver.js \| Quarantined ... |
| 13 | 2026-09-11 09:23:33 | quarantine | File: buynow_driver.js \| Quarantined At: 2026-... |
| 14 | 2026-09-11 09:23:33 | quarantine | File: edge_driver.js \| Quarantined At: 2026-09... |
| 15 | 2026-09-11 09:23:33 | quarantine | File: wallet-buynow.bundle.js \| Quarantined At... |
| 16 | 2026-09-11 09:23:33 | quarantine | File: app-setup.js \| Quarantined At: 2026-09-1... |
| 17 | 2026-09-11 09:23:33 | quarantine | File: bnpl.bundle.js \| Quarantined At: 2026-09... |
| 18 | 2026-09-11 09:23:33 | quarantine | File: notification.bundle.js \| Quarantined At:... |
| 19 | 2026-09-11 09:23:33 | quarantine | File: webui-setup.js \| Quarantined At: 2026-09... |
| 20 | 2026-09-11 09:23:33 | quarantine | File: miniwallet.bundle.js \| Quarantined At: 2... |
| 21 | 2026-09-11 09:35:09 | quarantine | File: bnpl.bundle.js \| Quarantined At: 2026-09... |
| 22 | 2026-09-11 09:35:09 | quarantine | File: app-setup.js \| Quarantined At: 2026-09-1... |
| 23 | 2026-09-11 09:35:09 | quarantine | File: bnpl_driver.js \| Quarantined At: 2026-09... |
| 24 | 2026-09-11 09:35:09 | quarantine | File: shopping_iframe_driver.js \| Quarantined ... |
| 25 | 2026-09-11 09:35:09 | quarantine | File: notification_fast.bundle.js \| Quarantine... |
| 26 | 2026-09-11 09:35:09 | quarantine | File: load-hub-i18n.bundle.js \| Quarantined At... |
| 27 | 2026-09-11 09:35:09 | quarantine | File: wallet_donation_driver.js \| Quarantined ... |
| 28 | 2026-09-11 09:35:09 | quarantine | File: edge_driver.js \| Quarantined At: 2026-09... |
| 29 | 2026-09-11 09:35:09 | quarantine | File: c3f9c4a416e9e493005fc3f97adacbcf12956a81... |
| 30 | 2026-09-11 09:35:09 | quarantine | File: webui-setup.js \| Quarantined At: 2026-09... |
| 31 | 2026-09-11 09:35:09 | quarantine | File: wallet-buynow.bundle.js \| Quarantined At... |
| 32 | 2026-09-11 09:35:09 | quarantine | File: notification.bundle.js \| Quarantined At:... |
| 33 | 2026-09-11 09:35:09 | quarantine | File: wallet_checkout_autofill_driver.js \| Qua... |
| 34 | 2026-09-11 09:35:09 | quarantine | File: crypto.bundle.js \| Quarantined At: 2026-... |
| 35 | 2026-09-11 09:35:09 | quarantine | File: runtime.bundle.js \| Quarantined At: 2026... |
| 36 | 2026-09-11 09:35:09 | quarantine | File: tokenized-card.bundle.js \| Quarantined A... |
| 37 | 2026-09-11 09:35:09 | quarantine | File: buynow_driver.js \| Quarantined At: 2026-... |
| 38 | 2026-09-11 09:35:09 | quarantine | File: load-ec-deps.bundle.js \| Quarantined At:... |
| 39 | 2026-09-11 09:35:09 | quarantine | File: load-ec-i18n.bundle.js \| Quarantined At:... |
| 40 | 2026-09-11 09:35:09 | quarantine | No suspicious file found by type or extension. |
| 41 | 2026-09-11 09:35:09 | quarantine | File: miniwallet.bundle.js \| Quarantined At: 2... |


Challenge_2_Analysis.ipynb
"delta_seconds",
"direction",
"message",
"destination_type"
EP4YIFHV web records within ±2 hours: 113
Direction distribution:
direction
AFTER 104
BEFORE 9
Name: count, dtype: int64
Complete EP4YIFHV ±2h timeline:
113 rows × 5 columns


Challenge_2_Analysis.ipynb
display(
ep4_critical_web[
"event_time",
"message",
"destination_type"
EP4YIFHV web records from 09:00 to 09:40: 9


Challenge_2_Analysis.ipynb
# Cell 69 — EPD24DBC USB activity
usb_epd = (
usb_data[
usb_data["endpoint_id"] == "EPD24DBC"
]
.copy()
.sort_values("event_time")
.reset_index(drop=True)
)
print(
"EPD24DBC USB events:",
len(usb_epd)
)
display(usb_epd)
EPD24DBC USB events: 4
# Cell 70 — Security correlation around EPD24DBC USB activity
usb_epd = usb_epd.copy()
usb_epd["event_time"] = pd.to_datetime(usb_epd["event_time"])
security_data["event_time"] = pd.to_datetime(



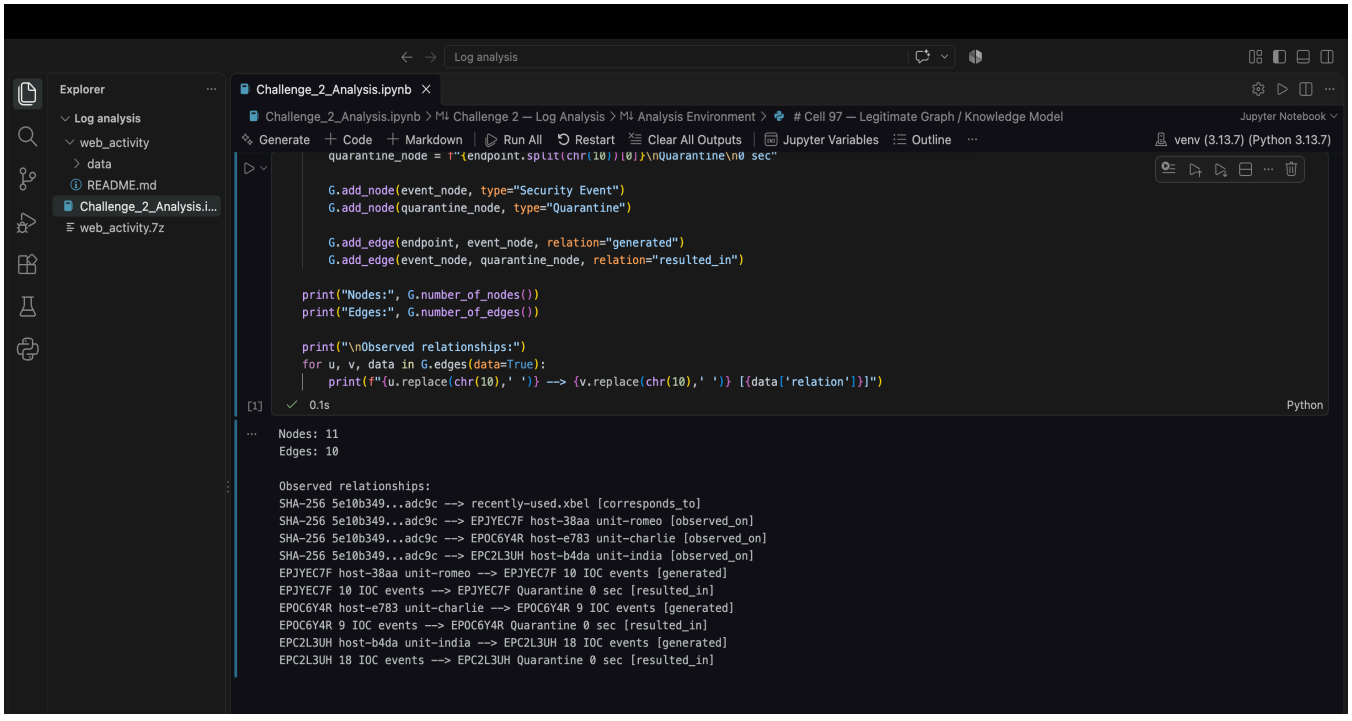
EPPGWIOF timestamp comparison:

| | event_time | malicious_events | malicious_files | security_events |
|---|---|---|---|---|
| 0 | 2026-09-01 08:01:19 | NaN | NaN | 2 |
| 1 | 2026-09-02 08:09:17 | NaN | NaN | 2 |
| 2 | 2026-09-03 08:03:25 | NaN | NaN | 2 |
| 3 | 2026-09-05 12:35:47 | NaN | NaN | 2 |
| 4 | 2026-09-06 10:02:20 | NaN | NaN | 2 |
| 5 | 2026-09-06 18:53:38 | NaN | NaN | 2 |
| 6 | 2026-09-07 09:24:25 | NaN | NaN | 2 |
| 7 | 2026-09-07 11:25:38 | 2.0 | brprintconflsr3, rawtobr3 | 3 |
| 8 | 2026-09-07 11:47:20 | 2.0 | brprintconflsr3, rawtobr3 | 3 |
| 9 | 2026-09-07 12:10:49 | 2.0 | brprintconflsr3, rawtobr3 | 3 |
| 10 | 2026-09-07 13:58:49 | 2.0 | brprintconflsr3, rawtobr3 | 3 |
| 11 | 2026-09-08 08:12:56 | 2.0 | brprintconflsr3, rawtobr3 | 3 |
| 12 | 2026-09-09 08:47:33 | NaN | NaN | 2 |
| 13 | 2026-09-09 18:54:25 | NaN | NaN | 2 |
| 14 | 2026-09-10 09:08:20 | NaN | NaN | 2 |
| 15 | 2026-09-10 18:34:48 | NaN | NaN | 2 |
| 16 | 2026-09-11 08:40:42 | NaN | NaN | 2 |
| 17 | 2026-09-11 12:52:40 | NaN | NaN | 2 |
| 18 | 2026-09-12 08:21:30 | NaN | NaN | 2 |
| 19 | 2026-09-13 09:36:51 | NaN | NaN | 2 |
| 20 | 2026-09-14 07:59:42 | NaN | NaN | 2 |
| 21 | 2026-09-14 17:15:18 | NaN | NaN | 2 |
| 22 | 2026-09-15 08:26:26 | NaN | NaN | 2 |
| 23 | 2026-09-16 08:15:58 | NaN | NaN | 2 |
| 24 | 2026-09-16 17:48:12 | NaN | NaN | 2 |
| 25 | 2026-09-17 08:10:26 | NaN | NaN | 2 |
| 26 | 2026-09-17 18:10:10 | NaN | NaN | 2 |
| 27 | 2026-09-18 08:08:52 | NaN | NaN | 2 |
| 28 | 2026-09-19 08:10:06 | NaN | NaN | 2 |
| 29 | 2026-09-21 08:07:39 | NaN | NaN | 2 |
| 30 | 2026-09-21 17:44:32 | NaN | NaN | 2 |
| 31 | 2026-09-22 08:02:39 | NaN | NaN | 2 |
| 32 | 2026-09-23 08:30:36 | NaN | NaN | 2 |
| 33 | 2026-09-24 08:16:08 | NaN | NaN | 2 |
| 34 | 2026-09-25 08:03:49 | NaN | NaN | 2 |
| 35 | 2026-09-26 07:34:09 | NaN | NaN | 2 |

## Step 12 — Graph relationship modelling

Purpose: Demonstrate graph/knowledge analytics using observed relationships.

**SCREENSHOT PLACEHOLDER — Step 12 — Graph relationship modelling**



```
        quarantine_node = f"{endpoint.split(chr(10))[0]}\nQuarantine\n0 sec"

        G.add_node(event_node, type="Security Event")
        G.add_node(quarantine_node, type="Quarantine")

        G.add_edge(endpoint, event_node, relation="generated")
        G.add_edge(event_node, quarantine_node, relation="resulted_in")

print("Nodes:", G.number_of_nodes())
print("Edges:", G.number_of_edges())

print("\nObserved relationships:")
for u, v, data in G.edges(data=True):
    print(f"{u.replace(chr(10),' ')} --> {v.replace(chr(10),' ')} [{data['relation']}]")
```

```
Nodes: 11
Edges: 10

Observed relationships:
SHA-256 5e10b349...adc9c --> recently-used.xbel [corresponds_to]
SHA-256 5e10b349...adc9c --> EPJYEC7F host-38aa unit-romeo [observed_on]
SHA-256 5e10b349...adc9c --> EPOC6Y4R host-e783 unit-charlie [observed_on]
SHA-256 5e10b349...adc9c --> EPC2L3UH host-b4da unit-india [observed_on]
EPJYEC7F host-38aa unit-romeo --> EPJYEC7F 10 IOC events [generated]
EPJYEC7F 10 IOC events --> EPJYEC7F Quarantine 0 sec [resulted_in]
EPOC6Y4R host-e783 unit-charlie --> EPOC6Y4R 9 IOC events [generated]
EPOC6Y4R 9 IOC events --> EPOC6Y4R Quarantine 0 sec [resulted_in]
EPC2L3UH host-b4da unit-india --> EPC2L3UH 18 IOC events [generated]
EPC2L3UH 18 IOC events --> EPC2L3UH Quarantine 0 sec [resulted_in]
```




## Step 13 — Final evidence matrix

Purpose: Consolidate the strongest evidence in one view.

### SCREENSHOT PLACEHOLDER — Step 13 — Final evidence matrix

Click to add a breakpoint ue)

```
display(final_evidence)
```

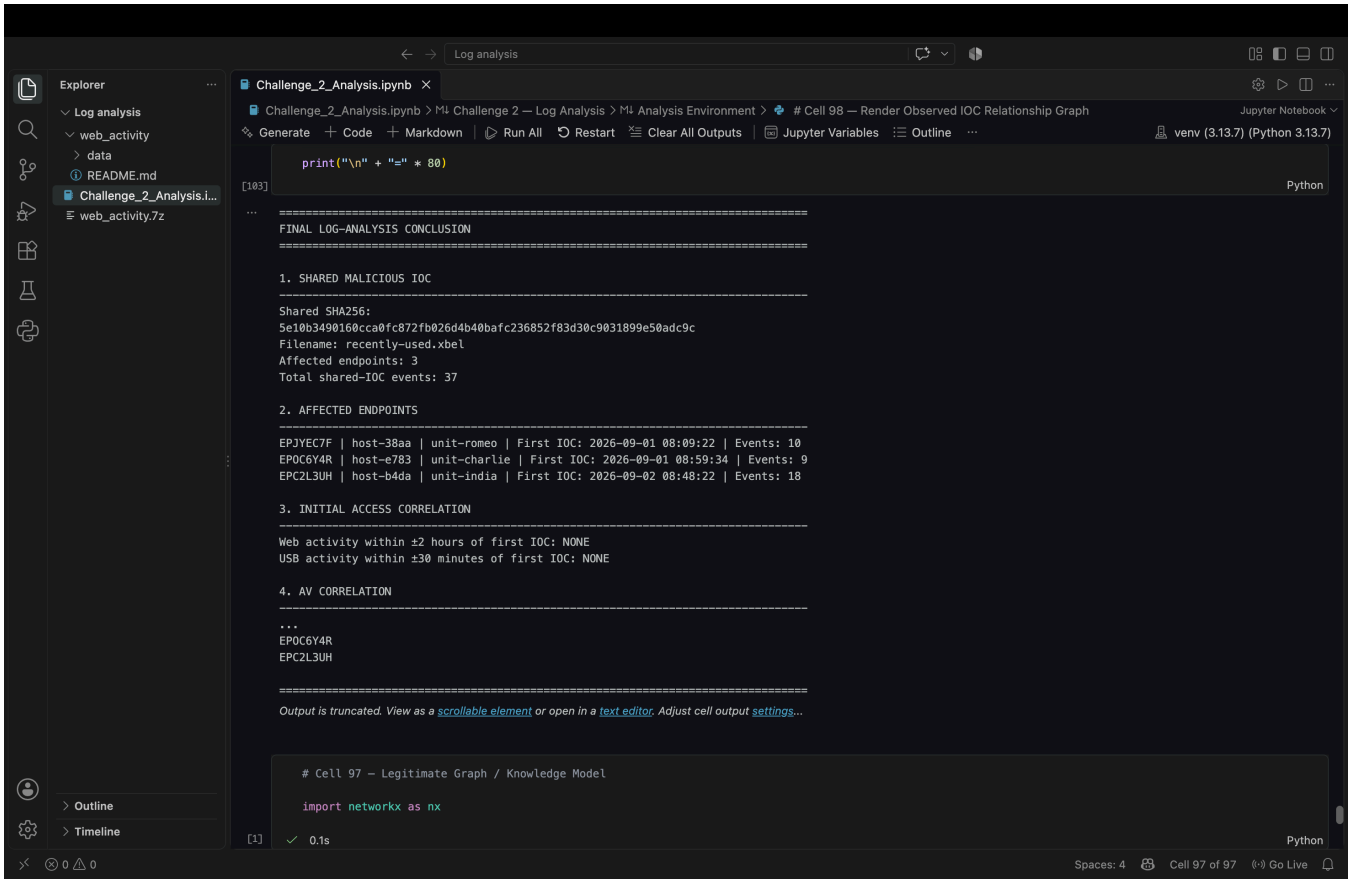
Python

| | endpoint_id | endpoint_name | unit | first_shared_ioc | last_shared_ioc | shared_ioc_events | web_events_±2h | usb_events_±30m | strict_av_events_±30m | first_ioc_quaran |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | EPJYEC7F | host-38aa | unit-romeo | 2026-09-01 08:09:22 | 2026-09-26 06:45:13 | 10 | 0 | 0 | 1 | |
| 1 | EPOC6Y4R | host-e783 | unit-charlie | 2026-09-01 08:59:34 | 2026-09-24 08:39:34 | 9 | 0 | 0 | 1 | |
| 2 | EPC2L3UH | host-b4da | unit-india | 2026-09-02 08:48:22 | 2026-09-23 08:45:44 | 18 | 0 | 0 | 1 | |

## Step 14 — Final conclusion

Purpose: Translate evidence into a defensible conclusion.

**SCREENSHOT PLACEHOLDER — Step 14 — Final conclusion**




# 6. Graph / Knowledge Analytics Layer

To align the analysis with the Graph and Knowledge Analytics role, the final graph should model only observed relationships. It must not represent a propagation or infection chain.

| Node / Edge | Observed relationship |
|---|---|
| **IOC → File** | The malicious SHA-256 corresponds to recently-used.xbel |
| **IOC → Endpoint** | The same IOC is observed on EPJYEC7F, EPOC6Y4R and EPC2L3UH |
| **Endpoint → Event** | Each endpoint has observed IOC events (10, 9 and 18) |

| | |
|---|---|
| **Event → Quarantine** | First observed IOC detection on each endpoint has immediate quarantine |

Important: EPJYEC7F → EPOC6Y4R → EPC2L3UH must not be drawn as a propagation path. It is only the chronological order of first observed detections.

**SCREENSHOT PLACEHOLDER — Graph / NetworkX result**

**ATTACH SCREENSHOT HERE**
**Reference: New final graph cell created after the existing investigation**

What to capture: Rendered graph containing IOC, recently-used.xbel, three endpoints and their observed event/quarantine relationships

Why: Demonstrate legitimate graph modelling without introducing unsupported causality

Expected/key result: One connected observed-evidence graph; no propagation arrows

# 7. Final Evidence Matrix

| Endpoint | Shared IOC | IOC Events | Web ±2h | USB ±30m | First IOC Quarantine | Assessment |
|---|---|---|---|---|---|---|
| EPJYEC7F | Yes | 10 | 0 | 0 | Yes (0 sec) | Associated/affected |
| EPOC6Y4R | Yes | 9 | 0 | 0 | Yes (0 sec) | Associated/affected |
| EPC2L3UH | Yes | 18 | 0 | 0 | Yes (0 sec) | Associated/affected |

# 8. Limitations and Forensic Cautions

- The logs establish artifact detection/association and quarantine, but do not establish execution or persistence.
- No matching web activity was found in the tested ±2-hour window; this does not prove that internet access was impossible.
- No matching USB activity was found in the tested ±30-minute window; this does not globally rule out USB as a source.
- The first-detection chronology is not proof of endpoint-to-endpoint propagation.
- Generic AV counts such as 2,622 av_scan events are not infection counts.

- The temporally correlated 1,673 security records are not 1,673 independent infections.
- Other endpoints such as EP3GALVS and EPR4BEDS were not added to the primary IOC list without exact SHA-256 verification.

## 9. Final Conclusion

Three endpoints — EPJYEC7F (unit-romeo), EPOC6Y4R (unit-charlie), and EPC2L3UH (unit-india) — are associated with the same malicious SHA-256 IOC 5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c, corresponding to recently-used.xbel. The IOC generated 37 observed events across the three endpoints.

The first detections occurred on 01 September 2026 at 08:09:22 and 08:59:34, and on 02 September 2026 at 08:48:22 respectively. Each endpoint's first observed IOC detection was accompanied by an immediate quarantine event at the same timestamp.

Correlation with available web activity within ±2 hours and USB activity within ±30 minutes of the first IOC detections produced no matching events. Therefore, the available telemetry does not establish the initial access vector.

The strongest supported finding is that the three endpoints were affected/associated with the same malicious artifact and that the artifact was detected and quarantined by the security controls. The available logs are insufficient to establish how the artifact initially entered the environment or whether one affected endpoint infected another.

## 10. Appendix

This reports is been done through using Python -Pandas in VS code and has approx. 100 cells of programme, therefore thre has approx. 100+ screenshots etc which is not necessary to put in this reports...

